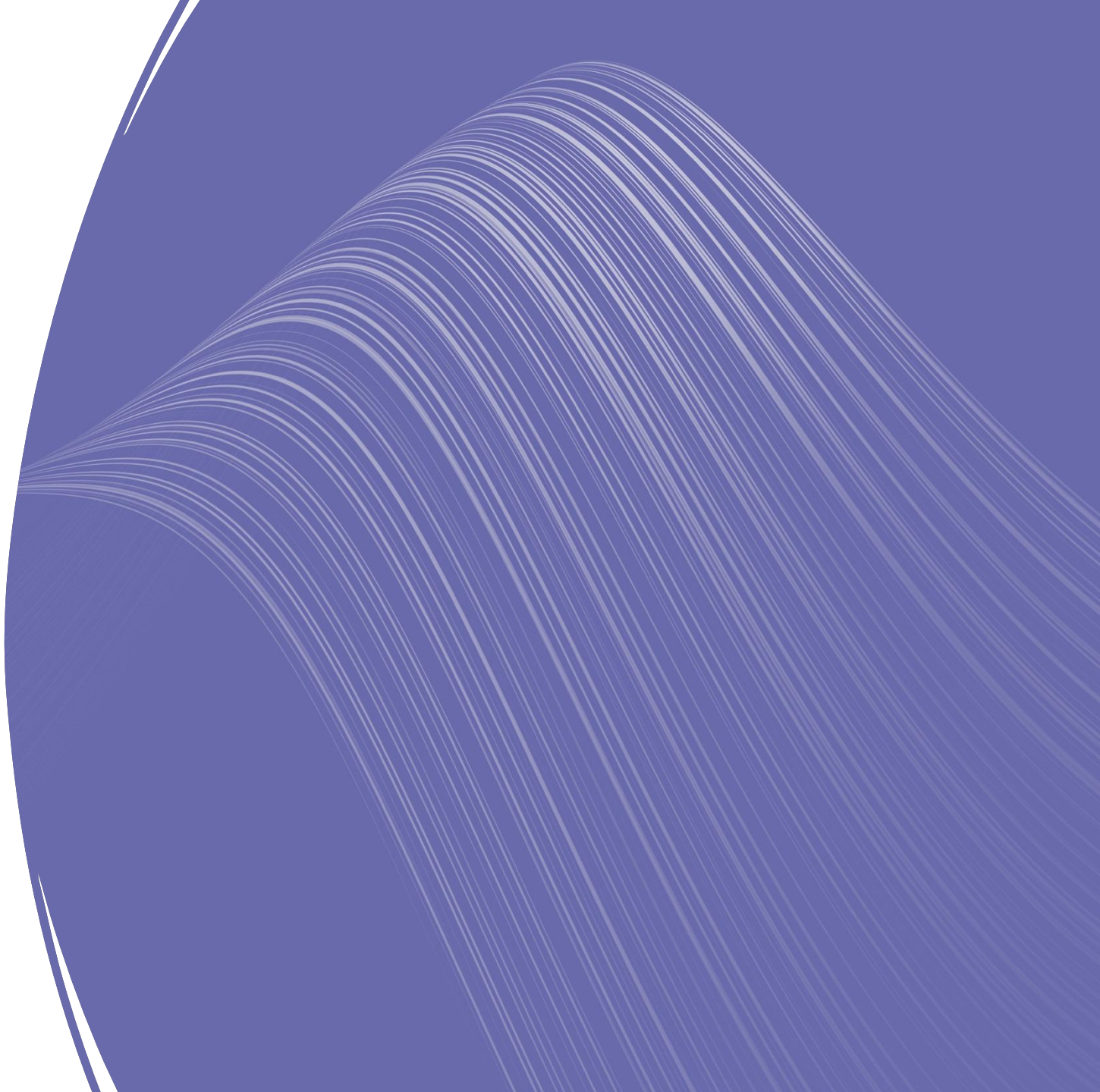
CITB106 & CSCB035 2026/2027

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СТРУКТУРА НА КУРСОВЕТЕ
ПОНЕДЕЛНИК 14:40 – 16:10

Лекции & Упражнения

Лекции първите 6 седмици (online):

1. Делимост - бройни системи, прости числа, НОД и НОК
2. Модулна аритметика - операции в остатъчното поле
3. Приложение на модулната аритметика при кодиране и защита на информация - хеш функции, афинни шифри, ISBN код.
4. Подмножества - преброяване, генериране и кодиране
5. Пермутации - преброяване, генериране и кодиране
6. Разбиване на числа и множества - преброяване, генериране и кодиране

Препоръчителна литература:

- Kenneth H. Rosen, **Discrete mathematics and its applications** (7th edition), 2013
- Кр. Манев, Увод в дискретната математика, НБУ, 1996; КЛМН, София, 2003
- C. Stein, R. Drysdale, K. Bogart, *Discrete mathematics for Computer Scientists*

СТРУКТУРА НА КУРСОВЕТЕ

Упражнения:

Във втората си част курсът е насочен към комбинаторни алгоритми за преброяване, генериране и кодиране на основни комбинаторни конфигурации.

Лекции & Упражнения

1. Шилдт Х., *Практически самоучител по* C, Софтпрес, 2001
2. cplusplus.com - *The C++ Resources Network* - http://www.cplusplus.com/
3. Eckel B., *Thinking in C++*, 2nd edition, v.1,v2, http://mindview.net/Books/TICPP/ThinkingInCPP2e.html

СТРУКТУРА НА КУРСОВЕТЕ

Текущ Контрол

- Първи текущ контрол (ноември, 2026)
- Втори текущ контрол (януари, 2027)

Текущият контрол ще се проведе под формата на online тестове (за първия времето за работа е 60 минути, а за втория - 90 минути), като тежестта им за формирането на крайна оценка от текущ контрол е в съотношение 1:2. При резултат от поне добър /4/ се освобождавате от финалния изпит.
Важно: При оценка Слаб /2/ на който и да е от двата текущи контроли е задължително да се явите на финалния изпит.

СТРУКТУРА НА КУРСОВЕТЕ

Текущ Контрол

Оценъчна скала за **ТЕОРЕТИЧНИЯ ИЗПИТ (първи текущ контрол)**:

0%-40% - Слаб(2.00)

41%-55% - Среден(3.00)

56%-70% - Добър(4.00)

71%-85% - Много Добър(5.00)

86%-100% - Отличен(6.00).

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Делимост

Частно и Остатък

Definition 1: If a and b are integers with a ≠ 0, we say that *a divides b* if there is an integer c such that b = ac, or equivalently, if $\frac{b}{a}$ is an integer. When a divides b we say that a is a *factor* or *divisor* of b, and that b is a *multiple* of a. The notation a | b denotes that a divides b. We write $a \nmid b$ when a does not divide b.

Theorem 1: Let a, b, and c be integers, where a ≠ 0. Then

(i) If a|b and a|c, then a|(b+c);

(ii) If a|b, then a|bc for allintegers c;

(iii) If a|b and b|c, then a|c.

Proof (iii): $b = ad; c = be \Rightarrow c = (ad)e = a(de) \Rightarrow a|c$

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Theorem 2 (The division algorithm): Let a be an integer and d a positive integer. Then there are unique integers q and r, with $0 \leq r < d$, such that $a = dq + r$.

Definition 2: In the equality given in the division algorithm, d is called the *divisor*, a is called the *dividend*, q is called the *quotient*, and r is called the *remainder*. This notation is used to express the quotient and remainder:

$$q = a \; div \; d; \quad r = a \; mod \; d$$

Делимост

Частно и Остатък

Examples:

100 = 11*9 + 1; 9 = 100 *div* 11; 1 = 100 *mod* 11

-13 = 3*(-5) +2; -5 = -13 *div* 3; 2 = -13 *mod* 3

Incorrect: -13 = 3*(-4) – 1 ???

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Modular

Arithmetic

Definition 3: If a and b are integers and m is a positive integer, then a is *congruent to b modulo m* if m divides a – b. We use the notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m.
We say that $a \equiv b \pmod{m}$ is a **congruence** and that m is its **modulus** (plural **moduli**).
If a and b are not congruent modulo m, we write $a \not\equiv b \pmod{m}$.

Theorem 3: Let a and b be integers, and let m be a positive integer.
Then $a \equiv b \pmod{m}$ if and only if a **mod** m = b **mod** m

Definition 4: The set of all integers congruent to an integer a modulo m is called the **congruence class** of a modulo m.

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Modular

Arithmetic

Theorem 4: Let m be a positive integer.
If a ≡ b (mod m) and c ≡ d (mod m),
then a + c ≡ b + d (mod m) and ac ≡ bd (mod m).

Proof:

$$b = a + pm; d = c + qm \Rightarrow bd = (a + pm)(c + qm) =$$
$$= ac + aqm + cpm + pqmm = ac + (aq + cp + pqm)m = ac + tm$$
$$\Rightarrow bd \equiv ac(mod\, m)$$

Caution: $ac \equiv bc(mod\, m) \Rightarrow a \equiv b(mod\, m)$ could be false!!!

Example: a = 6; b = 3; c = 11; d = 2; m = 3; ac = 66; bd = 6
3 | (6-3); 3 | (11-2) ⇒ 66 $\equiv 6\, mod\, 3$, or 3 | (66 – 6)

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Modular Arithmetic

We can define arithmetic operations on $\mathbf{Z_m}$, the set of nonnegative integers less than m, that is, the set $\{0, 1, \ldots, m - 1\}$. In particular, we define addition of these integers, denoted by $+_m$ by $a +_m b = (a + b)$ **mod** m, where the addition on the right-hand side of this equation is the ordinary addition of integers, and we define multiplication of these integers, denoted by $\cdot_m$ by $a \cdot_m b = (a \cdot b)$ **mod** m, where the multiplication on the right-hand side of this equation is the ordinary multiplication of integers. The operations $+_m$ and $\cdot_m$ are called addition and multiplication modulo m and when we use these operations, we are said to be doing **arithmetic modulo** m.

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

НОД (gcd)

Definition 5: Let a and b be integers, not both zero. The largest integer d such that d | a and d | b is called the *greatest common divisor* of a and b. The greatest common divisor of a and b is denoted by gcd(a, b) or (a, b).

Examples:

gcd(14, 4) = ???;

gcd(13, 6) = ???;

Definition 6: The integers a and b are *relatively prime* if their greatest common divisor is 1.

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

НОК (lcm)

Definition 7: The *least common multiple* of the positive integers a and b is the smallest positive integer that is divisible by both a and b. The least common multiple of a and b is denoted by lcm(a, b) or [a, b].

Examples:

lcm(14, 4) = ???;

lcm(13, 6) = ???;

Theorem 5: Let a and b be positive integers. Then $ab = \gcd(a, b) \cdot \mathrm{lcm}(a, b)$.

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Прости числа

Primes

Definition 8: An integer p greater than 1 is called *prime* if the only positive factors of p are 1 and p. A positive integer that is greater than 1 and is not prime is called *composite*.

Theorem 6 (The Fundamental Theorem of Arithmetic): Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

Trial Division: An integer is prime if it is not divisible by any prime less than or equal to its square root. This leads to the brute-force algorithm: we divide n by all primes not exceeding $\sqrt{n}$ and conclude that n is prime if it is not divisible by any of these primes.

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Прости числа

Primes

Theorem 7: There are infinitely many primes.

Proof: Use the Fundamental Theorem of Arithmetic and use proof by contradiction, i.e. assume that there are finitely many primes….

The Twin Prime Conjecture: Twin primes are pairs of primes that differ by 2, such as 3 and 5, 5 and 7, 11 and 13, 17 and 19, and 4967 and 4969. The twin prime conjecture asserts that there are infinitely many twin primes. One of the strongest results proved concerning twin primes is that there are infinitely many pairs p and p + 2, where p is prime and p + 2 is prime or the product of two primes (proved by J. R. Chen in 1966).

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Алгоритъм на Евклид

Theorem 8: Let a and b be positive integers. Then $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$.

Lemma 1: Let $a = bq + r$, where a, b, q, and r are integers.
Then $\gcd(a, b) = \gcd(b, r)$.

Proof: Suppose that d divides both a and b. Then it follows that d also divides $a - bq = r$ because $a = dt$ and $b = ds$ implies that $a - bq = dt - (ds)q = d(t - sq) = r$, i.e. $d \mid r$. Hence, any common divisor of a and b is also a common divisor of b and r.

Likewise, suppose that d divides both b and r. Then d also divides $bq + r = a$. Hence, any common divisor of b and r is also a common divisor of a and b. Thus, $(a, b) = (b, r)$.

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Suppose that a and b are positive integers with $a \geq b$. Let $r_0 = a$ and $r_1 = b$.

$$
\begin{aligned}
r_0 &= r_1 q_1 + r_2 && 0 \leq r_2 < r_1, \\
r_1 &= r_2 q_2 + r_3 && 0 \leq r_3 < r_2, \\
&\vdots \\
r_{n-2} &= r_{n-1} q_{n-1} + r_n && 0 \leq r_n < r_{n-1}, \\
r_{n-1} &= r_n q_n.
\end{aligned}
$$

Алгоритъм на Евклид

From Lemma 1: $(a, b) = (r_0, r_1) = (r_1, r_2) = \ldots = (r_n, 0) = r_n$

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Алгоритъм на Евклид

ALGORITHM 1 The Euclidean Algorithm.

procedure $gcd(a, b$: positive integers)
$x := a$
$y := b$
while $y \neq 0$
    $r := x$ **mod** y
    $x := y$
    $y := r$
return $x\{\gcd(a, b)$ is $x\}$

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Definition 9: If a and b are positive integers, then integers s and t such that gcd(a, b) = sa + tb are called *Bézout coefficients* of a and b (after Étienne Bézout, a French mathematician of the eighteenth century). Also, the equation gcd(a, b) = sa + tb is called *Bézout's identity*.

Theorem 9 (BÉZOUT'S THEOREM): If a and b are positive integers, then there exist integers s and t such that gcd(a, b) = sa + tb.

Алгоритъм на Евклид

$$
\begin{aligned}
r_0 &= r_1 q_1 + r_2 && 0 \le r_2 < r_1, \\
r_1 &= r_2 q_2 + r_3 && 0 \le r_3 < r_2, \\
&\vdots \\
r_{n-2} &= r_{n-1} q_{n-1} + r_n && 0 \le r_n < r_{n-1}, \\
r_{n-1} &= r_n q_n.
\end{aligned}
$$

ТЕМА 1. ДЕЛИМОСТ. ЧАСТНО И ОСТАТЪК. НОД И НОК

Lemma 2: If a, b, and c are positive integers such that gcd(a,b) = 1 and a | bc, then a | c.
Proof: Use Bézout's theorem.

Lemma 3: If p is a prime and $p \mid a_1 a_2 \cdots a_n$, where each a_i is an integer, then $p \mid a_i$ for some i.
Proof: Use **Lemma 2**

Алгоритъм на Евклид

Lmma 3 helps to prove the uniqueness part of **The Fundamental Theorem of Arithmetic.**

Theorem 10: Let m be a positive integer and let a, b, and c be integers. If $ac \equiv bc \pmod{m}$ and gcd(c,m) = 1, then $a \equiv b \pmod{m}$.
Proof: Use **Lemma 2**.